Tutorial 2: Maximum Likelihood and Least Squares

Likelihood interpretation, estimation, and regression

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Tutorial 2: from likelihood to learning objectives

This tutorial reinforces the main ideas from Week 2:

- interpret likelihood correctly;
- understand Gaussian variance estimation;
- connect least squares with its geometric meaning;
- understand how probability assumptions create different losses.

Task 1: likelihood interpretation and boundary cases

Bernoulli model ▪ 12 minutes

A coin follows a Bernoulli model:

$$p(x \mid \pi) = \pi^x (1-\pi)^{1-x}, \qquad x \in \{0, 1\}.$$

Consider two datasets:

$$A = (1, 1, 1, 1, 1), \qquad B = (1, 0, 1, 1, 0).$$

Questions

1. In a likelihood function, what is fixed and what changes?
2. Derive the MLE of π for datasets A and B.
3. Why does dataset A produce a boundary solution?

Task 2: Gaussian variance estimation

MLE and interpretation ▪ 15 minutes

Suppose

$$x_1, \ldots, x_n \sim \mathcal{N}(\mu, \sigma^2),$$

and the mean has already been estimated:

$$\hat{\mu} = \bar{x}.$$

Questions

1. Derive the MLE estimate of σ^2.
2. Explain what the residual term $(x_i - \bar{x})^2$ represents.
3. Explain why the unbiased sample variance uses $n - 1$ instead of n.

Task 3: least squares interpretation

Geometry and model understanding ▪ 15 minutes

Consider the least-squares solution $\hat{\boldsymbol{w}}$:

$$\hat{\boldsymbol{w}} = \arg\min_{\boldsymbol{w}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

Questions

1. What does the residual $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{w}}$ represent?
2. The solution satisfies
$$\boldsymbol{X}^\top \boldsymbol{r} = 0.$$
Explain this condition geometrically.
3. Why is the explicit inverse $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$ usually avoided in practice?
4. If two feature columns are almost identical, what problem may occur?

Optional challenge: changing the noise model

From probability assumptions to loss functions

In Gaussian regression, the noise assumption leads to squared error.
Now suppose the regression noise follows a Laplace distribution:

$$p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}) = \frac{1}{2b} \exp\left(-\frac{|y_i - \boldsymbol{w}^\top \boldsymbol{x}_i|}{b}\right).$$

Questions

1. Derive the negative log-likelihood objective.
2. Compare it with Gaussian least squares.
3. Explain why the choice of noise model affects outliers.

Tutorial 2 complete

Questions and discussion